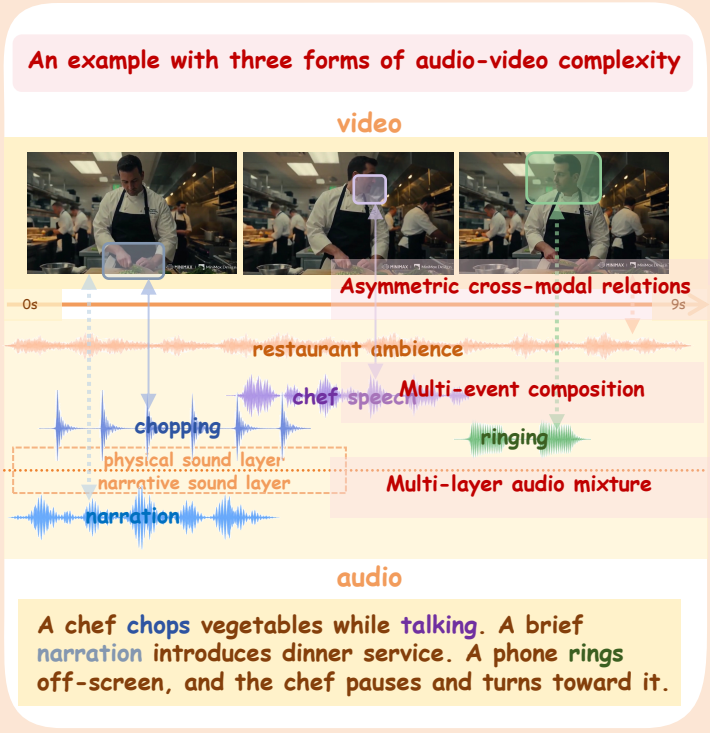
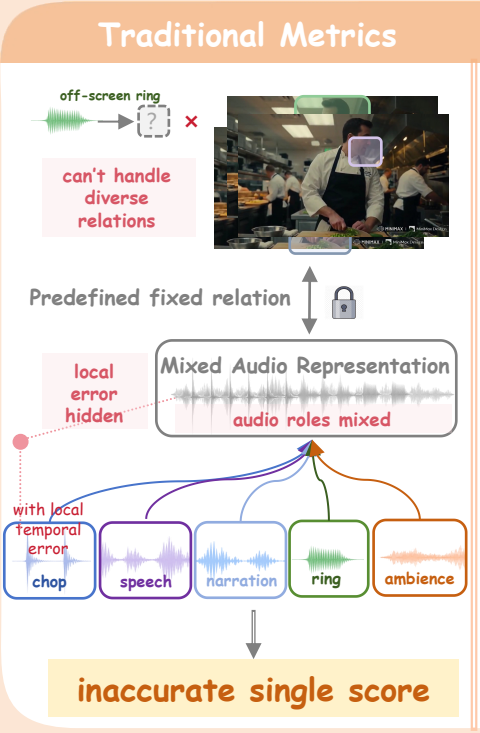
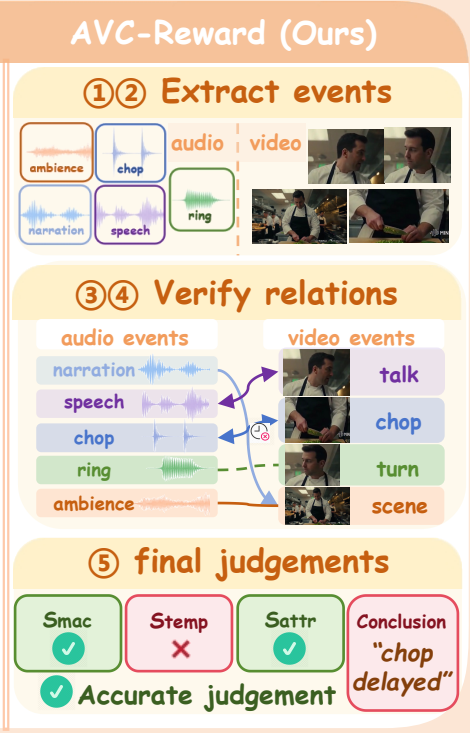

An example with three forms of audio-video complexity
video
Asymmetric cross-modal relations
0s
9s
restaurant ambience
chef speech
Multi-event composition
chopping
ringing
physical sound layer
narrative sound layer
Multi-layer audio mixture
narration
audio
A chef chops vegetables while talking. A brief narration introduces dinner service. A phone rings off-screen, and the chef pauses and turns toward it.


Traditional Metrics
off-screen ring
can't handle diverse relations
Predefined fixed relation
local error hidden
Mixed Audio Representation
audio roles mixed
with local temporal error
chop
speech
narration
ring
ambience
inaccurate single score


AVC-Reward (Ours)
①② Extract events
audio
video
ambience
chop
narration
speech
ring
③④ Verify relations
audio events
video events
narration
speech
chop
ring
ambience
talk
chop
turn
scene
⑤ final judgements
Smac
Stemp
Sattr
Conclusion
"chop delayed"
Accurate judgement